

EE24101: Basic Electrical Engineering

Module IV - Three-Phase Circuits

Text Book

- ▶ Fundamentals of Electric Circuits
- ▶ By *Charles K. Alexander and Matthew N. O. Sadiku*

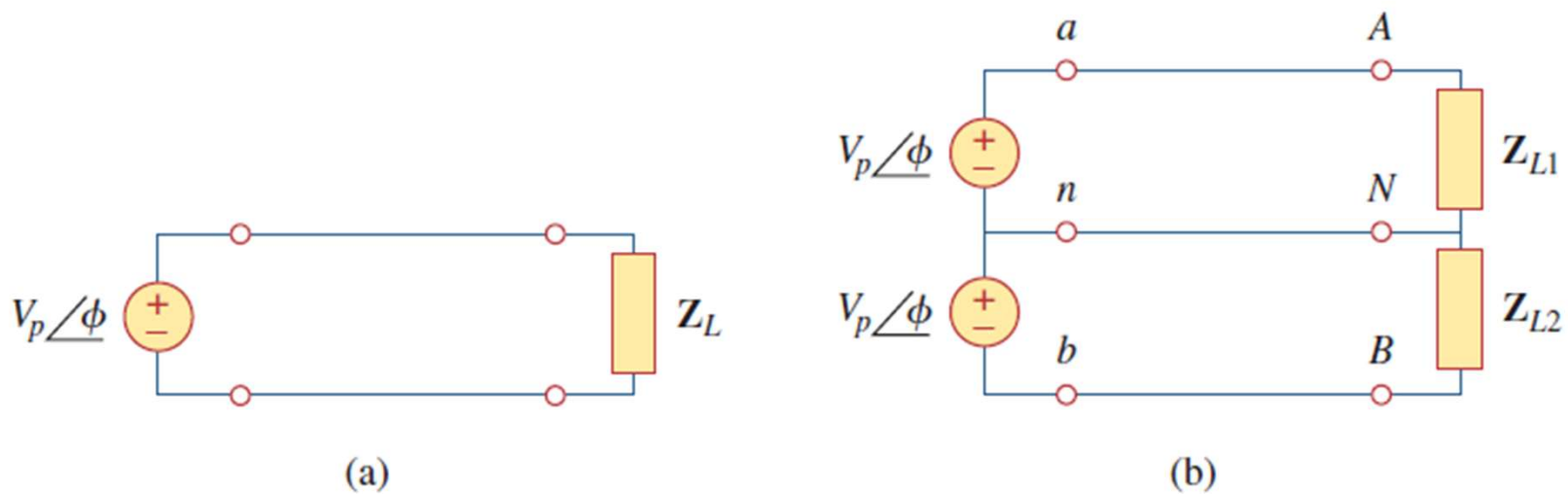


Figure 12.1

Single-phase systems: (a) two-wire type, (b) three-wire type.

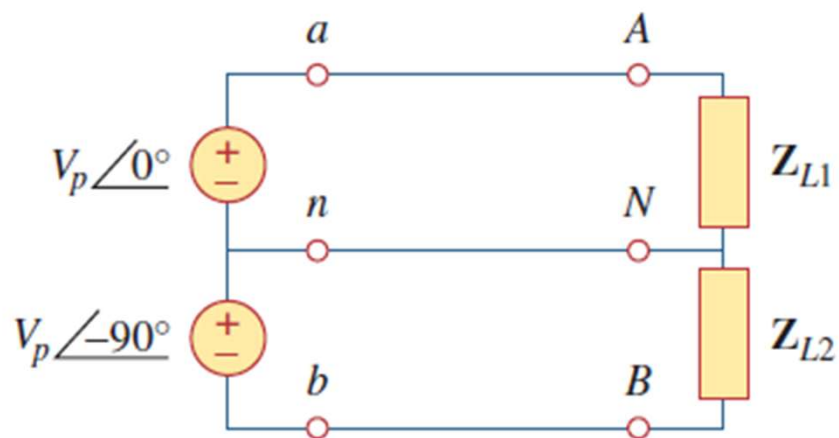


Figure 12.2

Two-phase three-wire system.

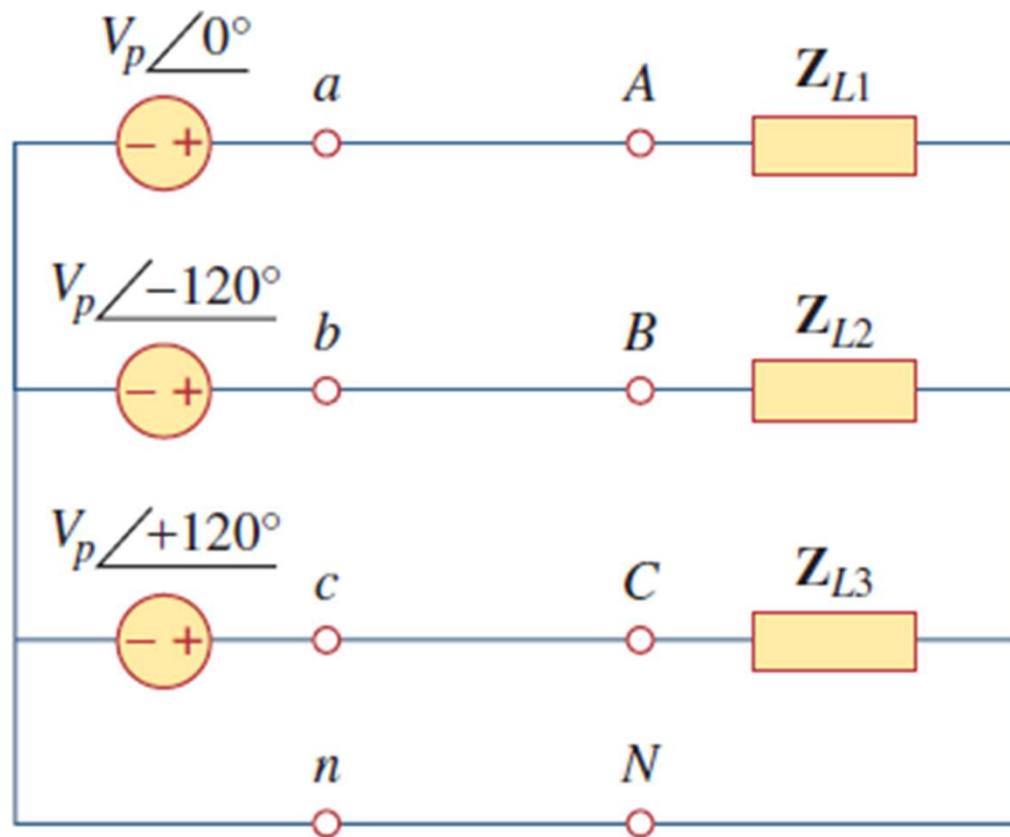


Figure 12.3
Three-phase four-wire system.

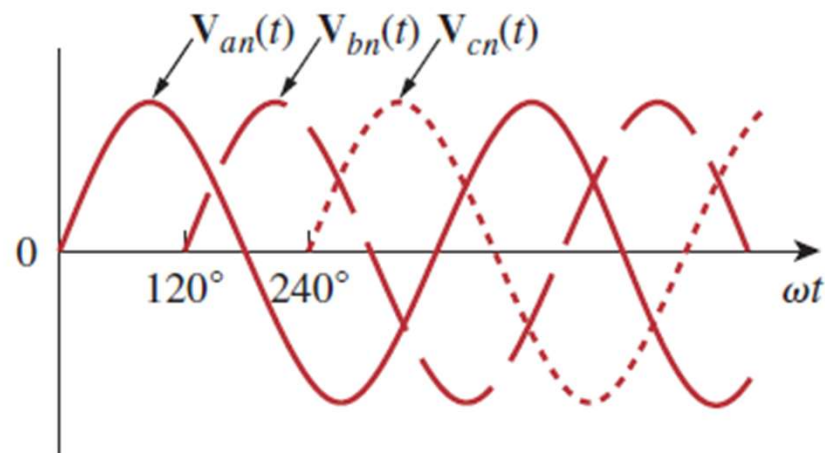
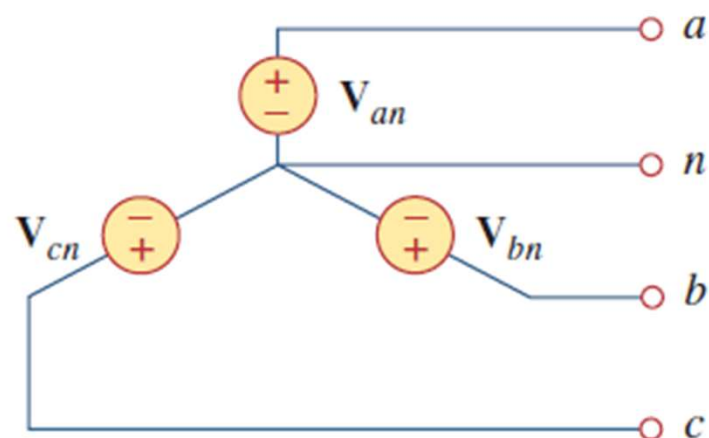
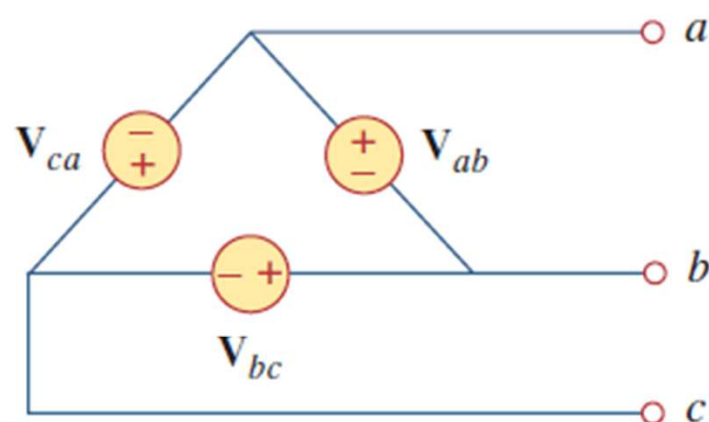


Figure 12.5

The generated voltages are 120° apart from each other.



(a)



(b)

Figure 12.6

Three-phase voltage sources: (a) Y-connected source, (b) Δ -connected source.

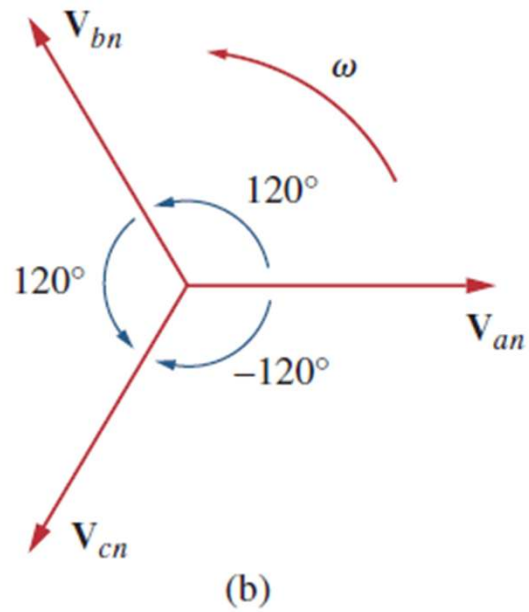
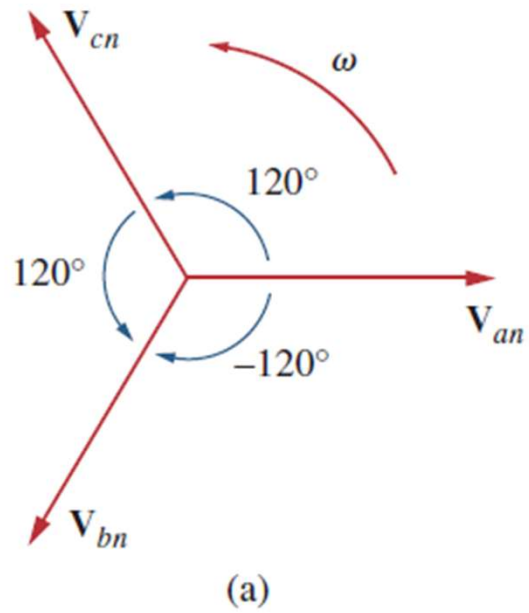


Figure 12.7

Phase sequences: (a) *abc* or positive sequence, (b) *acb* or negative sequence.

Let us consider the wye-connected voltages in Fig. 12.6(a) for now. The voltages $\mathbf{V}_{an}$, $\mathbf{V}_{bn}$, and $\mathbf{V}_{cn}$ are respectively between lines a , b , and c , and the neutral line n . These voltages are called *phase voltages*. If the voltage sources have the same amplitude and frequency ω and are out of phase with each other by 120° , the voltages are said to be *balanced*. This implies that

$$\mathbf{V}_{an} + \mathbf{V}_{bn} + \mathbf{V}_{cn} = 0 \quad (12.1)$$

$$|\mathbf{V}_{an}| = |\mathbf{V}_{bn}| = |\mathbf{V}_{cn}| \quad (12.2)$$

Thus,

Balanced phase voltages are equal in magnitude and are out of phase with each other by 120° .

Since the three-phase voltages are 120° out of phase with each other, there are two possible combinations. One possibility is shown in Fig. 12.7(a) and expressed mathematically as

$$\begin{aligned} \mathbf{V}_{an} &= V_p \angle 0^\circ \\ \mathbf{V}_{bn} &= V_p \angle -120^\circ \\ \mathbf{V}_{cn} &= V_p \angle -240^\circ = V_p \angle +120^\circ \end{aligned} \quad (12.3)$$

where V_p is the effective or rms value of the phase voltages. This is known as the *abc sequence* or *positive sequence*. In this phase sequence, V_{an} leads V_{bn} , which in turn leads V_{cn} . This sequence is produced when the rotor in Fig. 12.4 rotates counterclockwise. The other possibility is shown in Fig. 12.7(b) and is given by

$$\begin{aligned} V_{an} &= V_p \angle 0^\circ \\ V_{cn} &= V_p \angle -120^\circ \\ V_{bn} &= V_p \angle -240^\circ = V_p \angle +120^\circ \end{aligned} \quad (12.4)$$

This is called the *acb sequence* or *negative sequence*. For this phase sequence, V_{an} leads V_{cn} , which in turn leads V_{bn} . The *acb* sequence is produced when the rotor in Fig. 12.4 rotates in the clockwise direction. It is easy to show that the voltages in Eqs. (12.3) or (12.4) satisfy Eqs. (12.1) and (12.2). For example, from Eq. (12.3),

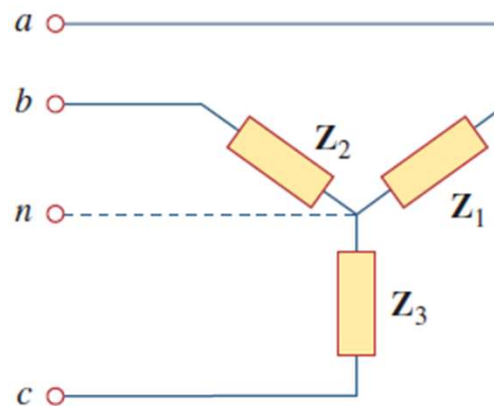
$$\begin{aligned} V_{an} + V_{bn} + V_{cn} &= V_p \angle 0^\circ + V_p \angle -120^\circ + V_p \angle +120^\circ \\ &= V_p(1.0 - 0.5 - j0.866 - 0.5 + j0.866) \quad (12.5) \\ &= 0 \end{aligned}$$

The **phase sequence** is the time order in which the voltages pass through their respective maximum values.

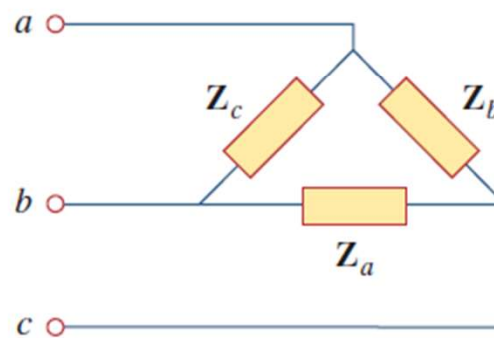
A **balanced load** is one in which the phase impedances are equal in magnitude and in phase.

For a *balanced* wye-connected load,

$$Z_1 = Z_2 = Z_3 = Z_Y \quad (12.6)$$



(a)



(b)

Figure 12.8

Two possible three-phase load configurations: (a) a Y-connected load, (b) a Δ -connected load.

where Z_Y is the load impedance per phase. For a *balanced* delta-connected load,

$$Z_a = Z_b = Z_c = Z_\Delta \quad (12.7)$$

where Z_Δ is the load impedance per phase in this case. We recall from Eq. (9.69) that

$$Z_\Delta = 3Z_Y \quad \text{or} \quad Z_Y = \frac{1}{3}Z_\Delta \quad (12.8)$$

so we know that a wye-connected load can be transformed into a delta-connected load, or vice versa, using Eq. (12.8).

Since both the three-phase source and the three-phase load can be either wye- or delta-connected, we have four possible connections:

- Y-Y connection (i.e., Y-connected source with a Y-connected load).
- Y- Δ connection.
- Δ - Δ connection.
- Δ -Y connection.

In subsequent sections, we will consider each of these possible configurations.

It is appropriate to mention here that a balanced delta-connected load is more common than a balanced wye-connected load. This is due to the ease with which loads may be added or removed from each phase of a delta-connected load. This is very difficult with a wye-connected load because the neutral may not be accessible. On the other hand, delta-connected sources are not common in practice because of the circulating current that will result in the delta-mesh if the three-phase voltages are slightly unbalanced.

Determine the phase sequence of the set of voltages

$$v_{an} = 200 \cos(\omega t + 10^\circ)$$

$$v_{bn} = 200 \cos(\omega t - 230^\circ), \quad v_{cn} = 200 \cos(\omega t - 110^\circ)$$

Solution:

The voltages can be expressed in phasor form as

$$\mathbf{V}_{an} = 200 \angle 10^\circ \text{ V}, \quad \mathbf{V}_{bn} = 200 \angle -230^\circ \text{ V}, \quad \mathbf{V}_{cn} = 200 \angle -110^\circ \text{ V}$$

We notice that $\mathbf{V}_{an}$ leads $\mathbf{V}_{cn}$ by 120° and $\mathbf{V}_{cn}$ in turn leads $\mathbf{V}_{bn}$ by 120° . Hence, we have an *acb* sequence.

12.3 Balanced Wye-Wye Connection

We begin with the Y-Y system, because any balanced three-phase system can be reduced to an equivalent Y-Y system. Therefore, analysis of this system should be regarded as the key to solving all balanced three-phase systems.

A **balanced Y-Y system** is a three-phase system with a balanced Y-connected source and a balanced Y-connected load.

Consider the balanced four-wire Y-Y system of Fig. 12.9, where a Y-connected load is connected to a Y-connected source. We assume a balanced load so that load impedances are equal. Although the impedance Z_Y is the total load impedance per phase, it may also be regarded as the sum of the source impedance Z_s , line impedance Z_ℓ , and load impedance Z_L for each phase, since these impedances are in series. As illustrated in Fig. 12.9, Z_s denotes the internal impedance of the phase winding of the generator; Z_ℓ is the impedance of the line joining a phase of the source with a phase of the load; Z_L is the impedance of each phase of the load; and Z_n is the impedance of the neutral line. Thus, in general

$$Z_Y = Z_s + Z_\ell + Z_L \quad (12.9)$$

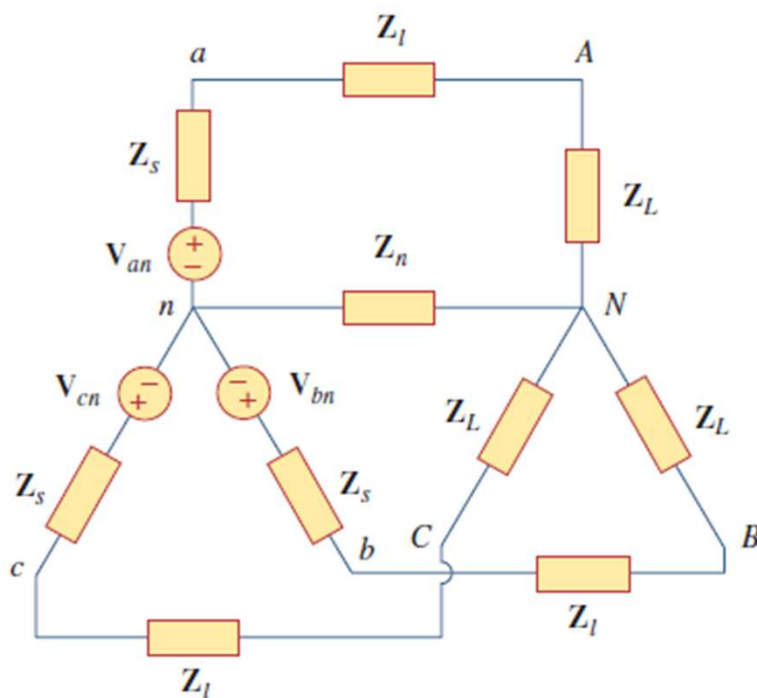


Figure 12.9

A balanced Y-Y system, showing the source, line, and load impedances.

Z_s and Z_ℓ are often very small compared with Z_L , so one can assume that $Z_Y = Z_L$ if no source or line impedance is given. In any event, by lumping the impedances together, the Y-Y system in Fig. 12.9 can be simplified to that shown in Fig. 12.10.

Assuming the positive sequence, the *phase* voltages (or line-to-neutral voltages) are

$$\begin{aligned} V_{an} &= V_p \angle 0^\circ \\ V_{bn} &= V_p \angle -120^\circ, \quad V_{cn} = V_p \angle +120^\circ \end{aligned} \quad (12.10)$$

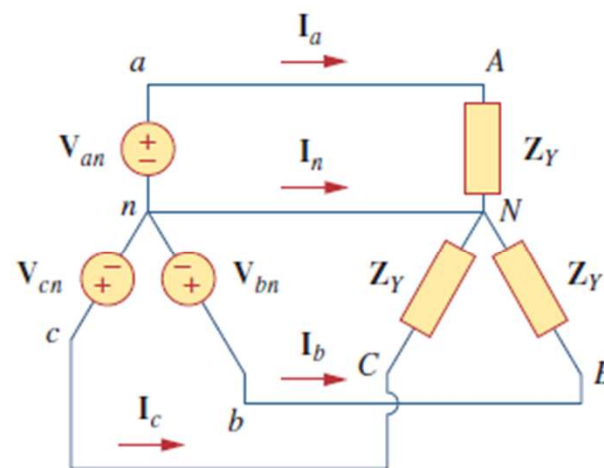


Figure 12.10

Balanced Y-Y connection.

The *line-to-line* voltages or simply *line* voltages V_{ab} , V_{bc} , and V_{ca} are related to the phase voltages. For example,

$$\begin{aligned} V_{ab} &= V_{an} + V_{nb} = V_{an} - V_{bn} = V_p \angle 0^\circ - V_p \angle -120^\circ \\ &= V_p \left(1 + \frac{1}{2} + j\frac{\sqrt{3}}{2} \right) = \sqrt{3} V_p \angle 30^\circ \end{aligned} \quad (12.11a)$$

Similarly, we can obtain

$$V_{bc} = V_{bn} - V_{cn} = \sqrt{3} V_p \angle -90^\circ \quad (12.11b)$$

$$V_{ca} = V_{cn} - V_{an} = \sqrt{3} V_p \angle -210^\circ \quad (12.11c)$$

Thus, the magnitude of the line voltages V_L is $\sqrt{3}$ times the magnitude of the phase voltages V_p , or

$$V_L = \sqrt{3} V_p$$

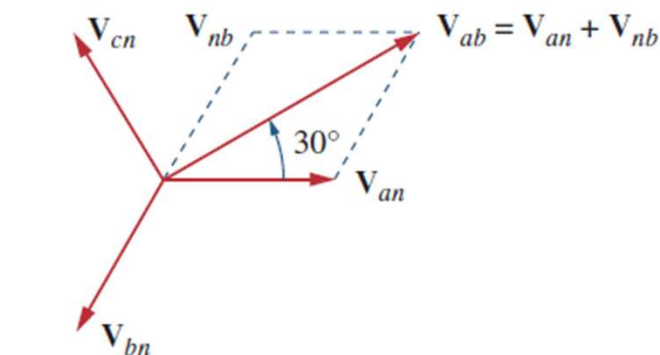
(12.12)

where

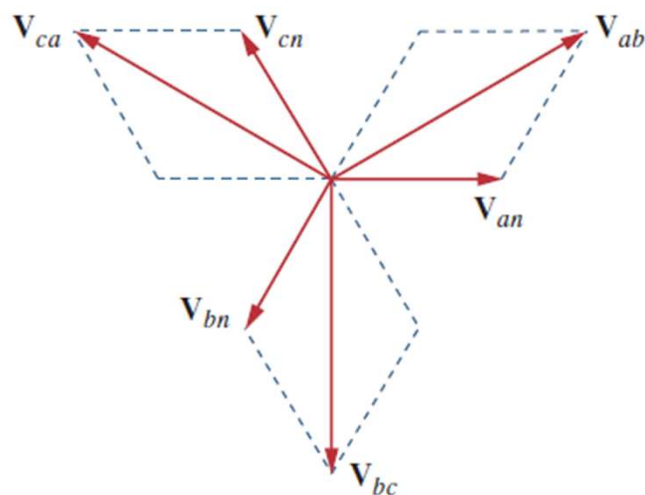
$$V_p = |V_{an}| = |V_{bn}| = |V_{cn}| \quad (12.13)$$

and

$$V_L = |V_{ab}| = |V_{bc}| = |V_{ca}| \quad (12.14)$$



(a)



(b)

Figure 12.11

Phasor diagrams illustrating the relationship between line voltages and phase voltages.

Also the line voltages lead their corresponding phase voltages by 30° . Figure 12.11(a) illustrates this. Figure 12.11(a) also shows how to determine V_{ab} from the phase voltages, while Fig. 12.11(b) shows the same for the three line voltages. Notice that V_{ab} leads V_{bc} by 120° , and V_{bc} leads V_{ca} by 120° , so that the line voltages sum up to zero as do the phase voltages.

Applying KVL to each phase in Fig. 12.10, we obtain the line currents as

$$\begin{aligned} \mathbf{I}_a &= \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y}, & \mathbf{I}_b &= \frac{\mathbf{V}_{bn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an} \angle -120^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a \angle -120^\circ \\ \mathbf{I}_c &= \frac{\mathbf{V}_{cn}}{\mathbf{Z}_Y} = \frac{\mathbf{V}_{an} \angle -240^\circ}{\mathbf{Z}_Y} = \mathbf{I}_a \angle -240^\circ \end{aligned} \quad (12.15)$$

We can readily infer that the line currents add up to zero,

$$\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0 \quad (12.16)$$

so that

$$\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) = 0 \quad (12.17a)$$

or

$$\mathbf{V}_{nN} = \mathbf{Z}_n \mathbf{I}_n = 0 \quad (12.17b)$$

that is, the voltage across the neutral wire is zero. The neutral line can thus be removed without affecting the system. In fact, in long distance power transmission, conductors in multiples of three are used with the earth itself acting as the neutral conductor. Power systems designed in this way are well grounded at all critical points to ensure safety.

While the *line* current is the current in each line, the *phase* current is the current in each phase of the source or load. In the Y-Y system, the line current is the same as the phase current. We will use single subscripts

for line currents because it is natural and conventional to assume that line currents flow from the source to the load.

An alternative way of analyzing a balanced Y-Y system is to do so on a “per phase” basis. We look at one phase, say phase a , and analyze the single-phase equivalent circuit in Fig. 12.12. The single-phase analysis yields the line current $\mathbf{I}_a$ as

$$\mathbf{I}_a = \frac{\mathbf{V}_{an}}{\mathbf{Z}_Y} \quad (12.18)$$

From $\mathbf{I}_a$, we use the phase sequence to obtain other line currents. Thus, as long as the system is balanced, we need only analyze one phase. We may do this even if the neutral line is absent, as in the three-wire system.

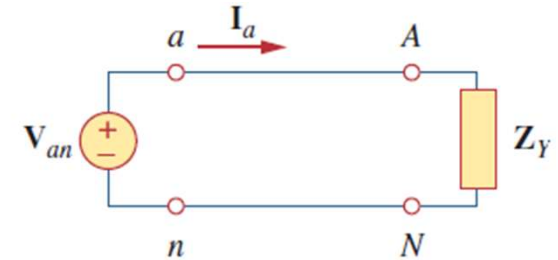


Figure 12.12
A single-phase equivalent circuit.

Calculate the line currents in the three-wire Y-Y system of Fig. 12.13.

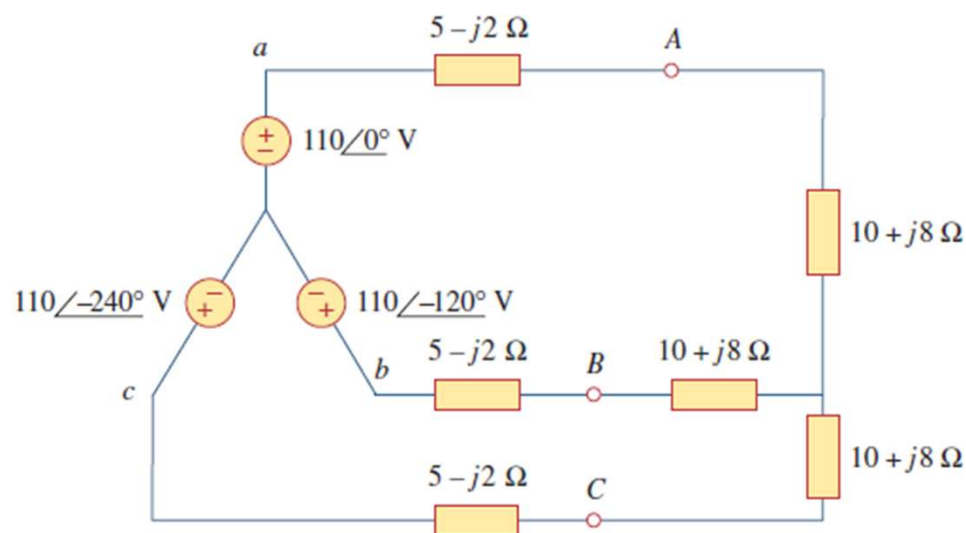


Figure 12.13

Three-wire Y-Y system; for Example 12.2.

Solution:

The three-phase circuit in Fig. 12.13 is balanced; we may replace it with its single-phase equivalent circuit such as in Fig. 12.12. We obtain I_a from the single-phase analysis as

$$I_a = \frac{V_{an}}{Z_Y}$$

where $Z_Y = (5 - j2) + (10 + j8) = 15 + j6 = 16.155 \angle 21.8^\circ$. Hence,

$$I_a = \frac{110 \angle 0^\circ}{16.155 \angle 21.8^\circ} = 6.81 \angle -21.8^\circ \text{ A}$$

Since the source voltages in Fig. 12.13 are in positive sequence, the line currents are also in positive sequence:

$$I_b = I_a \angle -120^\circ = 6.81 \angle -141.8^\circ \text{ A}$$

$$I_c = I_a \angle -240^\circ = 6.81 \angle -261.8^\circ \text{ A} = 6.81 \angle 98.2^\circ \text{ A}$$

12.4 Balanced Wye-Delta Connection

A **balanced Y-Δ system** consists of a balanced Y-connected source feeding a balanced Δ-connected load.

The balanced Y-delta system is shown in Fig. 12.14, where the source is Y-connected and the load is Δ-connected. There is, of course, no neutral connection from source to load for this case. Assuming the positive sequence, the phase voltages are again

$$\begin{aligned} V_{an} &= V_p \angle 0^\circ \\ V_{bn} &= V_p \angle -120^\circ, \quad V_{cn} = V_p \angle +120^\circ \end{aligned} \quad (12.19)$$

As shown in Section 12.3, the line voltages are

$$\begin{aligned} V_{ab} &= \sqrt{3}V_p \angle 30^\circ = V_{AB}, \quad V_{bc} = \sqrt{3}V_p \angle -90^\circ = V_{BC} \\ V_{ca} &= \sqrt{3}V_p \angle -150^\circ = V_{CA} \end{aligned} \quad (12.20)$$

showing that the line voltages are equal to the voltages across the load impedances for this system configuration. From these voltages, we can obtain the phase currents as

$$I_{AB} = \frac{V_{AB}}{Z_\Delta}, \quad I_{BC} = \frac{V_{BC}}{Z_\Delta}, \quad I_{CA} = \frac{V_{CA}}{Z_\Delta} \quad (12.21)$$

These currents have the same magnitude but are out of phase with each other by 120° .

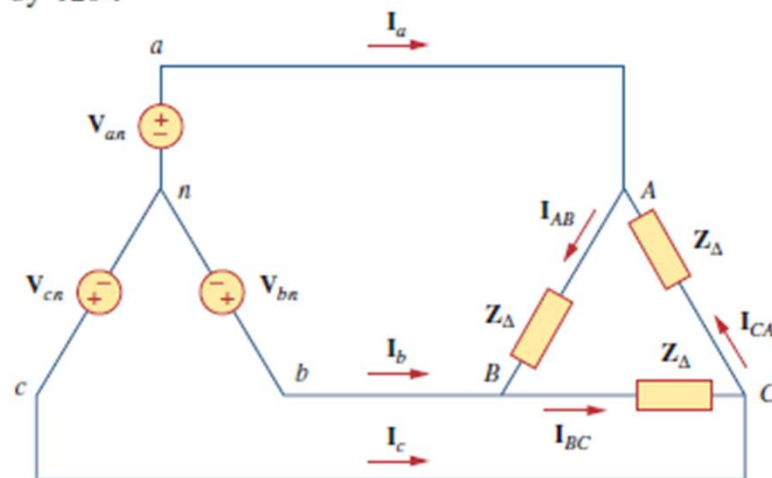


Figure 12.14
Balanced Y-Δ connection.

Another way to get these phase currents is to apply KVL. For example, applying KVL around loop $aABbna$ gives

$$-V_{an} + Z_{\Delta} I_{AB} + V_{bn} = 0$$

or

$$I_{AB} = \frac{V_{an} - V_{bn}}{Z_{\Delta}} = \frac{V_{ab}}{Z_{\Delta}} = \frac{V_{AB}}{Z_{\Delta}} \quad (12.22)$$

which is the same as Eq. (12.21). This is the more general way of finding the phase currents.

The line currents are obtained from the phase currents by applying KCL at nodes A , B , and C . Thus,

$$I_a = I_{AB} - I_{CA}, \quad I_b = I_{BC} - I_{AB}, \quad I_c = I_{CA} - I_{BC} \quad (12.23)$$

Since $I_{CA} = I_{AB} \angle -240^\circ$,

$$\begin{aligned} I_a &= I_{AB} - I_{CA} = I_{AB}(1 - 1 \angle -240^\circ) \\ &= I_{AB}(1 + 0.5 - j0.866) = I_{AB} \sqrt{3} \angle -30^\circ \end{aligned} \quad (12.24)$$

showing that the magnitude I_L of the line current is $\sqrt{3}$ times the magnitude I_p of the phase current, or

$$I_L = \sqrt{3} I_p$$

(12.25)

where

$$I_L = |I_a| = |I_b| = |I_c| \quad (12.26)$$

and

$$I_p = |I_{AB}| = |I_{BC}| = |I_{CA}| \quad (12.27)$$

Also, the line currents lag the corresponding phase currents by 30° , assuming the positive sequence. Figure 12.15 is a phasor diagram illustrating the relationship between the phase and line currents.

An alternative way of analyzing the Y- Δ circuit is to transform the Δ -connected load to an equivalent Y-connected load. Using the Δ -Y transformation formula in Eq. (12.8),

$$Z_Y = \frac{Z_\Delta}{3} \quad (12.28)$$

After this transformation, we now have a Y-Y system as in Fig. 12.10. The three-phase Y- Δ system in Fig. 12.14 can be replaced by the single-phase equivalent circuit in Fig. 12.16. This allows us to calculate only the line currents. The phase currents are obtained using Eq. (12.25) and utilizing the fact that each of the phase currents leads the corresponding line current by 30° .

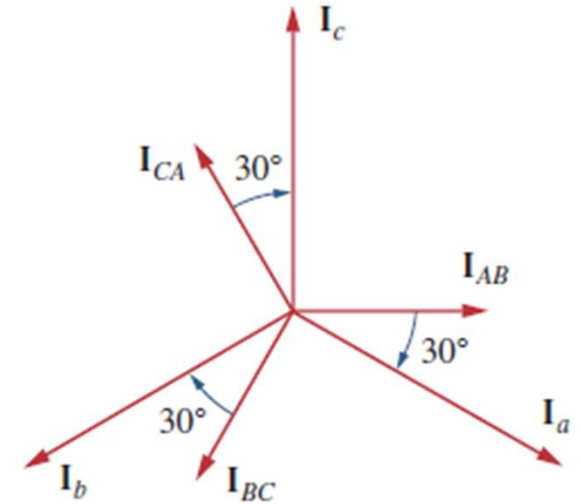


Figure 12.15

Phasor diagram illustrating the relationship between phase and line currents.

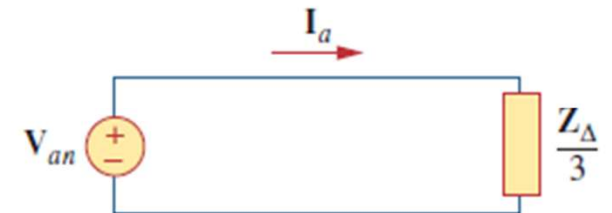


Figure 12.16

A single-phase equivalent circuit of a balanced Y- Δ circuit.

A balanced *abc*-sequence Y-connected source with $V_{an} = 100\angle 10^\circ$ V is connected to a Δ -connected balanced load $(8 + j4) \Omega$ per phase. Calculate the phase and line currents.

Solution:

This can be solved in two ways.

■ **METHOD 1** The load impedance is

$$Z_{\Delta} = 8 + j4 = 8.944\angle 26.57^\circ \Omega$$

If the phase voltage $V_{an} = 100\angle 10^\circ$, then the line voltage is

$$V_{ab} = V_{an}\sqrt{3}\angle 30^\circ = 100\sqrt{3}\angle 10^\circ + 30^\circ = V_{AB}$$

or

$$V_{AB} = 173.2\angle 40^\circ \text{ V}$$

The phase currents are

$$I_{AB} = \frac{V_{AB}}{Z_{\Delta}} = \frac{173.2\angle 40^\circ}{8.944\angle 26.57^\circ} = 19.36\angle 13.43^\circ \text{ A}$$

$$I_{BC} = I_{AB}\angle -120^\circ = 19.36\angle -106.57^\circ \text{ A}$$

$$I_{CA} = I_{AB}\angle +120^\circ = 19.36\angle 133.43^\circ \text{ A}$$

The line currents are

$$I_a = I_{AB}\sqrt{3}\angle -30^\circ = \sqrt{3}(19.36)\angle 13.43^\circ - 30^\circ = 33.53\angle -16.57^\circ \text{ A}$$

$$I_b = I_a\angle -120^\circ = 33.53\angle -136.57^\circ \text{ A}$$

$$I_c = I_a\angle +120^\circ = 33.53\angle 103.43^\circ \text{ A}$$

■ **METHOD 2** Alternatively, using single-phase analysis,

$$I_a = \frac{V_{an}}{Z_{\Delta}/3} = \frac{100\angle 10^\circ}{2.981\angle 26.57^\circ} = 33.54\angle -16.57^\circ \text{ A}$$

as above. Other line currents are obtained using the *abc* phase sequence.

12.7 Power in a Balanced System

Let us now consider the power in a balanced three-phase system. We begin by examining the instantaneous power absorbed by the load. This requires that the analysis be done in the time domain. For a Y-connected load, the phase voltages are

$$\begin{aligned}v_{AN} &= \sqrt{2}V_p \cos \omega t, & v_{BN} &= \sqrt{2}V_p \cos(\omega t - 120^\circ) \\v_{CN} &= \sqrt{2}V_p \cos(\omega t + 120^\circ)\end{aligned}\quad (12.41)$$

where the factor $\sqrt{2}$ is necessary because V_p has been defined as the rms value of the phase voltage. If $\mathbf{Z}_Y = Z \angle \theta$, the phase currents lag behind their corresponding phase voltages by θ . Thus,

$$\begin{aligned}i_a &= \sqrt{2}I_p \cos(\omega t - \theta), & i_b &= \sqrt{2}I_p \cos(\omega t - \theta - 120^\circ) \\i_c &= \sqrt{2}I_p \cos(\omega t - \theta + 120^\circ)\end{aligned}\quad (12.42)$$

where I_p is the rms value of the phase current. The total instantaneous power in the load is the sum of the instantaneous powers in the three phases; that is,

$$\begin{aligned} p &= p_a + p_b + p_c = v_{AN}i_a + v_{BN}i_b + v_{CN}i_c \\ &= 2V_p I_p [\cos \omega t \cos(\omega t - \theta) \\ &\quad + \cos(\omega t - 120^\circ) \cos(\omega t - \theta - 120^\circ) \\ &\quad + \cos(\omega t + 120^\circ) \cos(\omega t - \theta + 120^\circ)] \end{aligned} \quad (12.43)$$

Applying the trigonometric identity

$$\cos A \cos B = \frac{1}{2} [\cos(A + B) + \cos(A - B)] \quad (12.44)$$

gives

$$\begin{aligned} p &= V_p I_p [3 \cos \theta + \cos(2\omega t - \theta) + \cos(2\omega t - \theta - 240^\circ) \\ &\quad + \cos(2\omega t - \theta + 240^\circ)] \\ &= V_p I_p [3 \cos \theta + \cos \alpha + \cos \alpha \cos 240^\circ + \sin \alpha \sin 240^\circ \\ &\quad + \cos \alpha \cos 240^\circ - \sin \alpha \sin 240^\circ] \\ &\quad \text{where } \alpha = 2\omega t - \theta \end{aligned} \quad (12.45)$$

$$= V_p I_p \left[3 \cos \theta + \cos \alpha + 2 \left(-\frac{1}{2} \right) \cos \alpha \right] = 3 V_p I_p \cos \theta$$

Thus the total instantaneous power in a balanced three-phase system is constant—it does not change with time as the instantaneous power of each phase does. This result is true whether the load is Y- or Δ -connected. This is one important reason for using a three-phase system to generate and distribute power. We will look into another reason a little later.

Since the total instantaneous power is independent of time, the average power per phase P_p for either the Δ -connected load or the Y-connected load is $p/3$, or

$$P_p = V_p I_p \cos \theta \quad (12.46)$$

and the reactive power per phase is

$$Q_p = V_p I_p \sin \theta \quad (12.47)$$

The apparent power per phase is

$$S_p = V_p I_p \quad (12.48)$$

The complex power per phase is

$$S_p = P_p + jQ_p = V_p \mathbf{I}_p^* \quad (12.49)$$

where V_p and I_p are the phase voltage and phase current with magnitudes V_p and I_p , respectively. The total average power is the sum of the average powers in the phases:

$$P = P_a + P_b + P_c = 3P_p = 3V_p I_p \cos \theta = \sqrt{3} V_L I_L \cos \theta \quad (12.50)$$

For a Y-connected load, $I_L = I_p$ but $V_L = \sqrt{3}V_p$, whereas for a Δ -connected load, $I_L = \sqrt{3}I_p$ but $V_L = V_p$. Thus, Eq. (12.50) applies for both Y-connected and Δ -connected loads. Similarly, the total reactive power is

$$Q = 3V_p I_p \sin \theta = 3Q_p = \sqrt{3} V_L I_L \sin \theta \quad (12.51)$$

and the total complex power is

$$S = 3S_p = 3V_p \mathbf{I}_p^* = 3I_p^2 \mathbf{Z}_p = \frac{3V_p^2}{\mathbf{Z}_p^*} \quad (12.52)$$

where $\mathbf{Z}_p = Z_p \angle \theta$ is the load impedance per phase. ($\mathbf{Z}_p$ could be $\mathbf{Z}_Y$ or $\mathbf{Z}_\Delta$.) Alternatively, we may write Eq. (12.52) as

$$S = P + jQ = \sqrt{3}V_L I_L \angle \theta \quad (12.53)$$

Remember that V_p , I_p , V_L , and I_L are all rms values and that θ is the angle of the load impedance or the angle between the phase voltage and the phase current.

A second major advantage of three-phase systems for power distribution is that the three-phase system uses a lesser amount of wire than the single-phase system for the same line voltage V_L and the same absorbed power P_L . We will compare these cases and assume in both

12.8 † Unbalanced Three-Phase Systems

This chapter would be incomplete without mentioning unbalanced three-phase systems. An unbalanced system is caused by two possible situations: (1) The source voltages are not equal in magnitude and/or differ in phase by angles that are unequal, or (2) load impedances are unequal. Thus,

An **unbalanced system** is due to unbalanced voltage sources or an unbalanced load.

To simplify analysis, we will assume balanced source voltages, but an unbalanced load.

Unbalanced three-phase systems are solved by direct application of mesh and nodal analysis. Figure 12.23 shows an example of an unbalanced three-phase system that consists of balanced source voltages (not shown in the figure) and an unbalanced Y-connected load (shown in the figure). Since the load is unbalanced, Z_A , Z_B , and Z_C are not equal. The line currents are determined by Ohm's law as

$$I_a = \frac{V_{AN}}{Z_A}, \quad I_b = \frac{V_{BN}}{Z_B}, \quad I_c = \frac{V_{CN}}{Z_C} \quad (12.59)$$

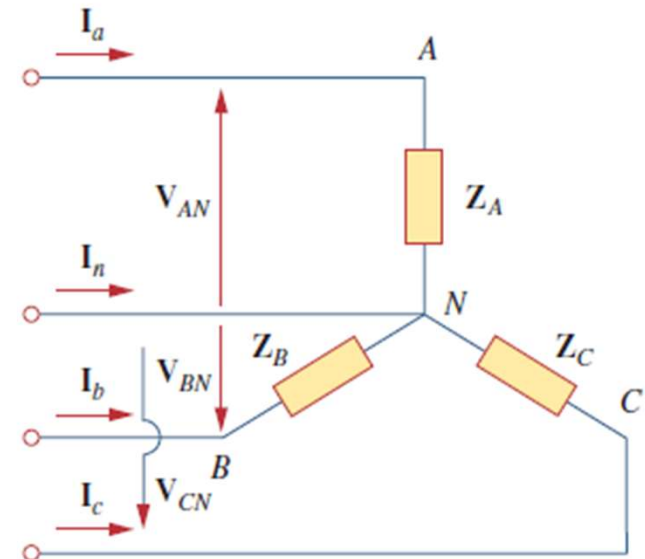


Figure 12.23

Unbalanced three-phase Y-connected load.

A special technique for handling unbalanced three-phase systems is the method of *symmetrical components*, which is beyond the scope of this text.

This set of unbalanced line currents produces current in the neutral line, which is not zero as in a balanced system. Applying KCL at node N gives the neutral line current as

$$\mathbf{I}_n = -(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c) \quad (12.60)$$

In a three-wire system where the neutral line is absent, we can still find the line currents $\mathbf{I}_a$, $\mathbf{I}_b$, and $\mathbf{I}_c$ using mesh analysis. At node N , KCL must be satisfied so that $\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$ in this case. The same could be done for an unbalanced Δ -Y, Y- Δ , or Δ - Δ three-wire system. As mentioned earlier, in long distance power transmission, conductors in multiples of three (multiple three-wire systems) are used, with the earth itself acting as the neutral conductor.

To calculate power in an unbalanced three-phase system requires that we find the power in each phase using Eqs. (12.46) to (12.49). The total power is not simply three times the power in one phase but the sum of the powers in the three phases.

12.10.1 Three-Phase Power Measurement

Section 11.9 presented the wattmeter as the instrument for measuring the average (or real) power in single-phase circuits. A single wattmeter can also measure the average power in a three-phase system that is balanced, so that $P_1 = P_2 = P_3$; the total power is three times the reading of that one wattmeter. However, two or three single-phase wattmeters are necessary to measure power if the system is unbalanced. The *three-wattmeter method* of power measurement, shown in Fig. 12.33, will work regardless of whether the load is balanced or unbalanced, wye- or delta-connected. The three-wattmeter method is well suited for power measurement in a three-phase system where the power factor is constantly changing. The total average power is the algebraic sum of the three wattmeter readings,

$$P_T = P_1 + P_2 + P_3 \quad (12.61)$$

where P_1 , P_2 , and P_3 correspond to the readings of wattmeters W_1 , W_2 , and W_3 , respectively. Notice that the common or reference point o in Fig. 12.33 is selected arbitrarily. If the load is wye-connected, point o can be connected to the neutral point n . For a delta-connected load, point o can be connected to any point. If point o is connected to point b , for example, the voltage coil in wattmeter W_2 reads zero and $P_2 = 0$, indicating that wattmeter W_2 is not necessary. Thus, two wattmeters are sufficient to measure the total power.

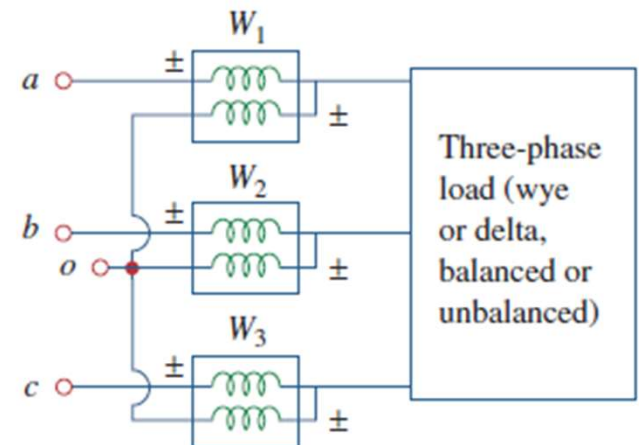


Figure 12.33

Three-wattmeter method for measuring three-phase power.

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The *two-wattmeter method* is the most commonly used method for three-phase power measurement. The two wattmeters must be properly connected to any two phases, as shown typically in Fig. 12.34. Notice that the current coil of each wattmeter measures the line current, while the respective voltage coil is connected between the line and the third line and measures the line voltage. Also notice that the $\pm$ terminal of the voltage coil is connected to the line to which the corresponding current coil is connected. Although the individual wattmeters no longer read the power taken by any particular phase, the algebraic sum of the two wattmeter readings equals the total average power absorbed by the load, regardless of whether it is wye- or delta-connected, balanced or

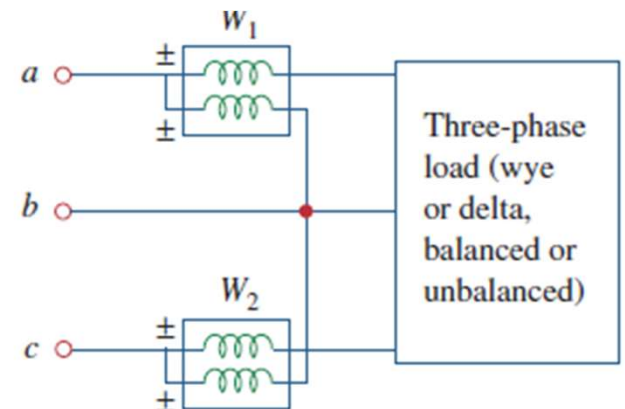


Figure 12.34

Two-wattmeter method for measuring three-phase power.

unbalanced. The total real power is equal to the algebraic sum of the two wattmeter readings,

$$P_T = P_1 + P_2 \quad (12.62)$$

We will show here that the method works for a balanced three-phase system.

Consider the balanced, wye-connected load in Fig. 12.35. Our objective is to apply the two-wattmeter method to find the average power absorbed by the load. Assume the source is in the abc sequence and the load impedance $Z_Y = Z_Y/\theta$. Due to the load impedance, each voltage coil leads its current coil by θ , so that the power factor is $\cos\theta$. We recall that each line voltage leads the corresponding phase voltage by 30° . Thus, the total phase difference between the phase current I_a and line voltage V_{ab} is $\theta + 30^\circ$, and the average power read by wattmeter W_1 is

$$P_1 = \text{Re}[V_{ab} I_a^*] = V_{ab} I_a \cos(\theta + 30^\circ) = V_L I_L \cos(\theta + 30^\circ) \quad (12.63)$$

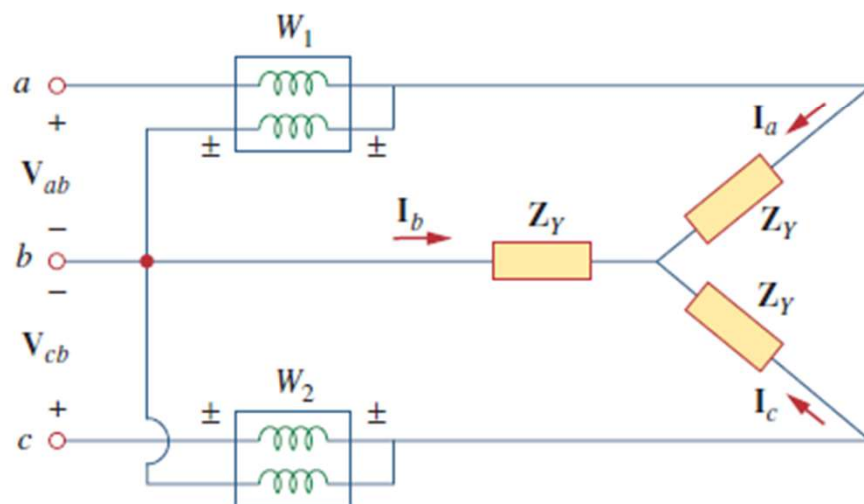


Figure 12.35

Two-wattmeter method applied to a balanced wye load.

Similarly, we can show that the average power read by wattmeter 2 is

$$P_2 = \text{Re}[V_{cb} \mathbf{I}_c^*] = V_{cb} I_c \cos(\theta - 30^\circ) = V_L I_L \cos(\theta - 30^\circ) \quad (12.64)$$

We now use the trigonometric identities

$$\begin{aligned} \cos(A + B) &= \cos A \cos B - \sin A \sin B \\ \cos(A - B) &= \cos A \cos B + \sin A \sin B \end{aligned} \quad (12.65)$$

to find the sum and the difference of the two wattmeter readings in Eqs. (12.63) and (12.64):

$$\begin{aligned} P_1 + P_2 &= V_L I_L [\cos(\theta + 30^\circ) + \cos(\theta - 30^\circ)] \\ &= V_L I_L (\cos \theta \cos 30^\circ - \sin \theta \sin 30^\circ \\ &\quad + \cos \theta \cos 30^\circ + \sin \theta \sin 30^\circ) \\ &= V_L I_L 2 \cos 30^\circ \cos \theta = \sqrt{3} V_L I_L \cos \theta \end{aligned} \quad (12.66)$$

since $2 \cos 30^\circ = \sqrt{3}$. Comparing Eq. (12.66) with Eq. (12.50) shows that the sum of the wattmeter readings gives the total average power,

$$P_T = P_1 + P_2$$

(12.67)

Similarly,

$$\begin{aligned}P_1 - P_2 &= V_L I_L [\cos(\theta + 30^\circ) - \cos(\theta - 30^\circ)] \\&= V_L I_L (\cos \theta \cos 30^\circ - \sin \theta \sin 30^\circ \\&\quad - \cos \theta \cos 30^\circ - \sin \theta \sin 30^\circ) \quad (12.68) \\&= -V_L I_L 2 \sin 30^\circ \sin \theta \\P_2 - P_1 &= V_L I_L \sin \theta\end{aligned}$$

since $2 \sin 30^\circ = 1$. Comparing Eq. (12.68) with Eq. (12.51) shows that the difference of the wattmeter readings is proportional to the total reactive power, or

$$Q_T = \sqrt{3}(P_2 - P_1)$$

 (12.69)

From Eqs. (12.67) and (12.69), the total apparent power can be obtained as

$$S_T = \sqrt{P_T^2 + Q_T^2} \quad (12.70)$$

Dividing Eq. (12.69) by Eq. (12.67) gives the tangent of the power factor angle as

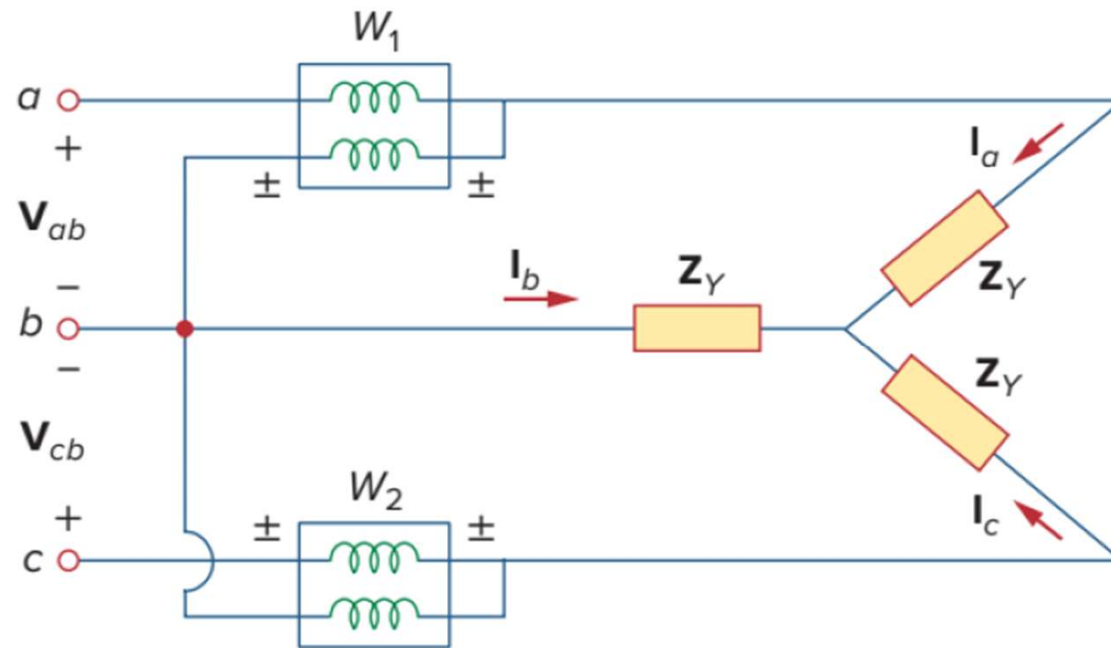
$$\tan \theta = \frac{Q_T}{P_T} = \sqrt{3} \frac{P_2 - P_1}{P_2 + P_1} \quad (12.71)$$

from which we can obtain the power factor as $\text{pf} = \cos\theta$. Thus, the two-wattmeter method not only provides the total real and reactive powers, it can also be used to compute the power factor. From Eqs. (12.67), (12.69), and (12.71), we conclude that:

1. If $P_2 = P_1$, the load is resistive.
2. If $P_2 > P_1$, the load is inductive.
3. If $P_2 < P_1$, the load is capacitive.

Although these results are derived from a balanced wye-connected load, they are equally valid for a balanced delta-connected load. However, the two-wattmeter method cannot be used for power measurement in a three-phase four-wire system unless the current through the neutral line is zero. We use the three-wattmeter method to measure the real power in a three-phase four-wire system.

The three-phase balanced load in figure, has impedance per phase of $Z_Y = 8 + j6 \, \Omega$. If the load is connected to 208-V lines, predict the readings of the wattmeter W_1 and W_2 . Find P_T and Q_T .



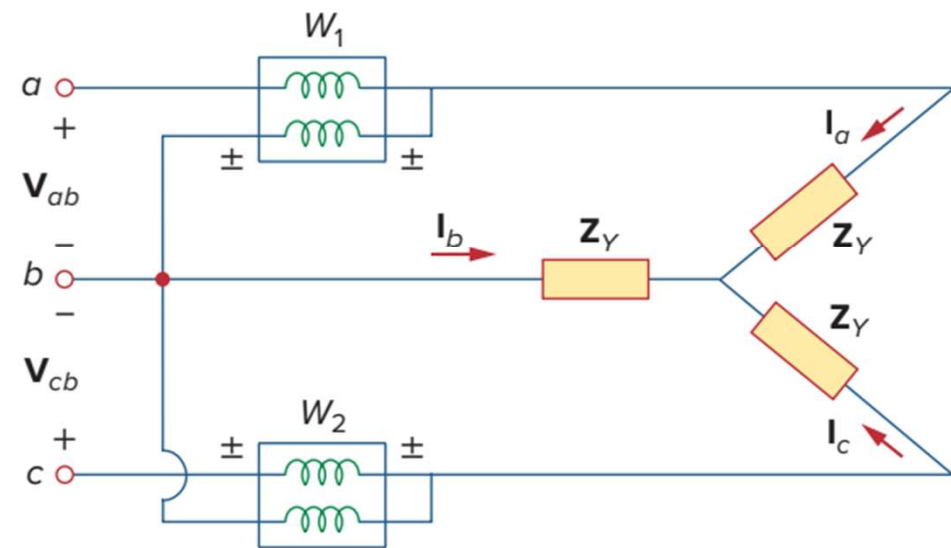
Solution:

The impedance per phase is $Z_Y = (8 + j6) = 10\angle 36.87^\circ$

Thus the Power Factor angle $\theta = 36.87^\circ$.

Since the line voltage $V_L = 208\text{ V}$,
the line current is

$$I_L = \frac{V_p}{|Z_Y|} = \frac{208/\sqrt{3}}{10} = 12\text{ A}$$



$$P_1 = V_L I_L \cos(\theta + 30^\circ) = 208 * 12 * \cos(36.87^\circ + 30^\circ) = 980.48W$$

$$P_2 = V_L I_L \cos(\theta - 30^\circ) = 208 * 12 * \cos(36.87^\circ - 30^\circ) = 2478.1W$$

Since $P_2 > P_1$, the load is inductive. This is evident from the load Z_Y itself.

$$P_T = P_1 + P_2 = 3.459kW$$

$$Q_T = \sqrt{3}(P_2 - P_1) = \sqrt{3}(1497.6)VAR = 2.594kVAR$$